Integrated Sensing and Communications for UAV Assisted Internet of Things Based on Deep Reinforcement Learning

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***Abstract*—As an emerging technology, Integrated Sensing and Communications (ISAC) is expected to be applied to uncrewed aerial vehicle (UAV)-assisted Internet of Things (IoT) systems to sense and acquire information from IoT devices in complex environments. In this paper, we design a UAV-ISAC assisted IoT system, where the UAV senses the IoT nodes in unknown locations and simultaneously forwards the sensing information to the ground base station (BS). The UAV is required to complete the sensing of all IoT nodes before returning to the starting position. However, several challenges arise in designing this system, including unknown IoT node positions, a non-convex optimization problem, constrained resources, a dynamic environment, and complex resource allocation. We propose using radar estimation rate as the measure for evaluating the UAV sensing performance. We formulate a UAV trajectory optimization problem that aims to maximize the radar estimation rate and minimize the UAV flight time, while adhering to constraints on radar sensing performance, information rate, and sensing all the nodes. However, solving this problem using standard optimization techniques is challenging due to the lack of information about the locations of IoT nodes and the non-convex nature of the problem. To address this challenge, we propose an algorithm by adopting multi-step learning Dueling Double Deep Q-learning network (DDQN) based on deep reinforcement learning (DRL) to solve this intractable optimization problem. The simulation results demonstrate that our proposed algorithm achieves stable convergence under different constraint thresholds. Furthermore, we verify the superiority of our algorithm by comparing it with other benchmark algorithms.**

***Index Terms*—Uncrewed aerial vehicle (UAV), Internet of Things (IoT), integrated sensing and communications (ISAC), deep reinforcement learning (DRL), radar estimation rate.**

I. Introduction

As WIRELESS communication frequency bands move to higher frequencies, they may overlap with traditional radar sensing frequency bands. Next-generation wireless communication systems aim to improve spectrum utilization by reusing radar frequency bands for both radar and communication purposes, which has led to the development of a technology called Integrated Sensing and Communication (ISAC). ISAC enables communication and sensing in the same frequency bands by sharing some or even the same set of devices [1], [2], [3]. In recent years, there has been a significant research interest in the design and application scenarios of ISAC [4], [5], [6], [7], [8]. One of the fields that has received substantial attention is the Internet of Things (IoT), as the two fundamental functions of sensing and communication are integral to the design and implementation of IoT [9], [10], [11].

Internet of Things (IoT) is constituted by the intelligent interconnections of various communication devices, such as sensors and routers, enabling nodes in the network to sense and transmit data to each other [12]. However, the channels between traditional ground IoT nodes generally suffer from significant path loss caused by factors such as shadowing and multipath fading. By leveraging the flexible deployment, wide-area coverage, and line-of-sight (LoS) link features of UAVs, the combination of UAVs and IoT, i.e., UAV-assisted IoT, can alleviate the above problems and lead to the advantage of expanding the communication range of IoT [13], [14], [15]. In the UAV assisted IoT system, the UAV needs to sense the IoT state and maintain connectivity with the IoT, which requires loading both communication and sensing devices. However, it is a challenge for UAVs to install communication system and sensing system independently due to their great power consumption. Fortunately, such a problem can be effectively solved by using UAV-enabled ISAC (UAV-ISAC). The UAV-ISAC assisted IoT system can provide ubiquitous communication and sensing functions, and enable efficient task execution to meet various requirements in complex scenarios, such as disaster response, temporary traffic surveillance, target tracking, and agricultural plant protection [16]. However, optimizing resources to improve the ISAC performance has become a critical challenge in the UAV-ISAC assisted IoT systems.

In recent years, many studies have been devoted to the UAV-ISAC system. In [17], with the goal of minimizing the total transmit power of the UAV-ISAC system, the subcarrier power allocations for ISAC and base station (BS) were cooperatively designed. [18] studied the resource allocation problem of joint communication and radar sensing in a UAV assisted network system based on rate split multiple access (RSMA). The deployment location and beam formation matrix of the UAV were

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jointly optimized under the constraints of power budget, transmit rate, and approximation error. In [19], the authors proposed a joint optimization problem of UAV localization, user association and UAV transmit power in order to balance communication and sensing performance. The designed resource optimization algorithm aims to increase the utility of a multi-UAV network with dual-function radar and communication capabilities.

Previous studies have primarily focused on the applications of UAVs in static scenarios, without fully leveraging the advantages of UAVs' mobility and controllability. The following literature studies specifically focus on the resource optimization of UAV-ISAC in maneuvering scenarios. In [20], the authors considered the joint design of UAV maneuvers (deployment position or flight trajectory) and transmit beamforming to maximize the average weighted sum throughput, while ensuring the sensing beampattern gain requirements, transmit power constraints, and practical flight constraints. In [21], the UAV-ISAC system sensed multiple targets through airborne radar equipment and transmitted the data to a ground BS for decision making. To guarantee the peak age of information (PAoI) for the UAV-ISAC system, the authors proposed a joint optimization problem that takes into account the UAV trajectory, scheduling order of the targets, and time and power allocation for packet sensing and communication. In [22], the authors aimed to maximize the user achievable rate by jointly optimizing the UAV trajectory, transmit precoder, and sensing start instant, subject to the constraints of sensing frequency and beam pattern gain. In [23], the authors aimed to maximize the achievable rate of the UAV-ISAC system by jointly optimizing UAV trajectory, user association, target sensing selection, and transmit beamforming, while ensuring that the sensing frequency and beammap gain requirements for a given target are met. In [26], the authors studied the problem of secrecy and energy efficiency for the UAV-ISAC system. To maximize the secrecy energy efficiency of the system, they proposed a joint optimization problem of UAV position, sensing time, and transmission time. In [27], the authors investigate a UAV-aided integrated sensing and mobile edge computing (MEC) system, aiming to reduce overall system energy consumption and data collection time through optimized sensing scheduling, power allocation, and UAV trajectory planning.

However, the above-mentioned literature has primarily studied the resource optimization for UAV-ISAC systems using traditional convex optimization techniques, which require full prior knowledge about the system and may have difficulty in solving strong non-convex optimization problems. In contrast, [24], [25] have turned to deep reinforcement learning (DRL) to address the challenges of complex resource optimization in UAV-ISAC systems. In [24], a novel reward combining Mutual Information (MI) and Communication Rate (CR) is defined for solving resource allocation problems in radar-communication integrated systems with a cluster of UAVs. The authors explore the implementation of DQN and dueling DQN algorithms using this new reward function. However, the authors appear to overlook the optimization and distribution of communication and sensing resources within the UAV-ISAC system framework. This oversight might limit the applicability and robustness of their methodology in UAV-ISAC environments. In [25], multiple UAVs operate as mobile aerial ISAC platforms, actively participating in both sensing and communication tasks with terrestrial target users. The authors formulate a joint optimization problem, encompassing user association, UAV trajectory planning, and power allocation, with the goal of maximizing the minimum weighted spectrum efficiency among the UAVs. However, narrowly focusing on maximizing this minimum weighted spectrum efficiency does not necessarily guarantee enhanced performance for either the communication or the sensing. Moreover, [24], [25] omit a pivotal consideration: in dynamic UAV-ISAC systems, given the finite energy reserves of UAVs, the constraint or optimization of UAV flight time becomes paramount. The comparison of related work with our work is shown in Table I.

In this paper, the UAV-ISAC simultaneously detects the IoT nodes and transmits the sensing information to the BS. We jointly optimize the radar sensing performance and flight time of the UAV-ISAC assisted IoT system by designing UAV flight trajectory, and propose a solution algorithm based on multi-step learning Dueling Double Deep Q-learning network (DDQN). The main contributions of this paper are summarized as follows

- We have developed a UAV-ISAC assisted IoT system, where a UAV serves as an aerial platform equipped with communication and sensing capabilities. The UAV-ISAC is designed to detect the IoT nodes in unknown locations and transmit the sensed information to a BS. We introduce radar estimation rate to measure the UAV radar sensing performance. Considering the energy constraint of the UAV, we propose an optimization problem that jointly maximizes system radar estimation rate and minimizes UAV flight time, subjected to the constraints of radar sensing performance, information rate and sensing all nodes.
- We transform the optimization problem into a Markov decision process (MDP), and then propose a DRL based solution algorithm to solve the optimization problem. We set up reward functions to enhance the agent's ability to learn effective strategies. Additionally, we utilize the advanced multi-step learning Dueling DDQN to achieve stable and faster convergence of the proposed algorithm.
- To verify the performance of the proposed algorithm, we conduct simulations and obtain results. The simulation results clearly demonstrate that the optimized UAV trajectory can effectively improve the system radar estimation rate and reduce the UAV flight time. Furthermore, based on the simulation results under various requirements for radar estimation rate, we can conclude that the algorithm exhibits good stability. In addition, the superiority of the proposed algorithm can be verified by the simulation results compared with the direct flight, DDQN, Greedy, Position-based and Random algorithms.

The rest of this paper is organized as follows. In Section II, we propose a UAV-ISAC assisted IoT system model. In Section III, we introduce radar estimation rate to measure the UAV sensing performance, and present a UAV trajectory optimization problem to jointly maximize system radar estimation rate and minimize UAV flight time. In Section IV, we provide DRL-based solution algorithm to solve the optimization problem. The simulation results and discussions are presented in Section V. Finally, the conclusions of the paper are drawn in Section VI.

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TABLE I
COMPARISON AMONG THE RELATED WORKS AND OUR WORK

Publications	Objective	Technique	Dynamic UAV	Connections of radar and communication	UAV fight time
[17]	Minimize the total transmit power consumption	Convex optimization	×	Independent	×
[19]	Maximize the total network utility	Convex optimization	×	Independent	×
[20]	Maximize the weighted sum-rate throughput	Convex optimization	✓	Independent	×
[21]	Minimize the maximum average PAoI	Convex optimization	✓	Independent	×
[22]	Maximize the user achievable rate	Convex optimization	✓	Independent	×
[23]	Maximize the achievable rate	Convex optimization	✓	Independent	×
[24]	Maximize the weight sum of MI and communication rate	DRL	✓	Independent	×
[25]	Maximize the minimum weighted spectral efficiency	DRL	✓	Independent	×
Our work	Maximize the radar estimation rate and minimize the UAV flight time	DRL	✓	Non independent	✓

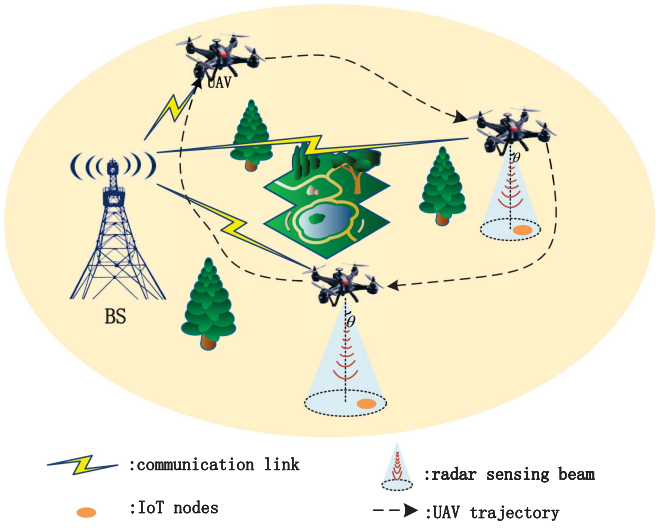


Fig. 1. UAV-ISAC assisted IoT system.

II. System Model

A. Scenario Description

As shown in Fig. 1, we consider a UAV-ISAC assisted IoT system consisting of a BS, a UAV with ISAC function and M IoT nodes. In the three-dimensional Cartesian coordinate system, the IoT nodes with the set of $\mathcal{M} = \{1, 2, \ldots, M\}$ are randomly distributed in an area $\mathcal{F}$, and the position of mth node is denoted by $\mathbf{g}_m = [x_m, y_m, 0], m \in \mathcal{M}$. It is worth noting that the locations of IoT nodes are unknown to the UAV. During the UAV flight over the area $\mathcal{F}$, the UAV-ISAC senses the IoT nodes and then sends the sensed information to the BS.

The position of the BS is denoted as $\mathbf{g} = [x_I, y_I, H_{BS}]$, where H_{BS} indicates the height of the BS. Furthermore, the instantaneous UAV position is denoted as $\mathbf{q}(t) = [x(t), y(t), H]$. For the convenience of problem modeling, we ignore the take-off and landing of the UAV, and fix its flight height at a constant altitude H. The UAV's flight altitude should be high enough to maintain the LoS link between the UAV and the IoT nodes. The UAV flies from the BS position with a fixed speed V, thus, the initial position of the UAV is denoted as $\mathbf{q}_I = [x_I, y_I, H]$. As the locations of the IoT nodes are unknown, it is impossible to determine whether there is an IoT node within the radar sensing coverage of the UAV by directly obtaining the distance between UAV and IoT node. We assume that the IoT nodes send their status signals to the UAV periodically, and the UAV evaluates whether an IoT node is within the sensing coverage based on the power strength of the received signal. If a node is within the sensing coverage, the UAV will simultaneously sense the IoT node and send the sensed information to the BS by splitting the power. The channel conditions between the UAV and the BS are usually poor. Therefore, if there is no node within the sensing coverage, the UAV will continue to transmit the remaining sensed information to the BS. When the UAV completes the sensing task of M IoT nodes, it will fly back to the starting position $\mathbf{q}_I$. The flight time of the UAV is denoted as T, which should be minimized to decrease the energy consumption.

B. Channel Model

We assume that the UAV and IoT nodes are equipped with half-wave dipole antennas, and the UAV is also equipped with separated receive antennas for receiving radar echoes. The BS uses an 8-element uniform linear array antenna (ULA) placed vertically in accordance with 3GPP specifications [28]. Furthermore, the BS site contains 3 sectors/cells with standard sectorization techniques. To optimize the BS coverage, pre-determined phase shifts are applied to the vertical antenna array so that its main lobe is electrically inclined to the ground by ϑ_{etilt} degree. Moreover, the distances between the antenna elements in the vertical and horizontal are set as $d_V = \frac{\lambda}{2}$ and $d_H = \frac{\lambda}{2}$, respectively, where λ represents the signal wavelength. The element radiation pattern $A_E(\vartheta, \phi)$ is composed of horizontal radiation pattern $A_{E,H}(\phi)$ and vertical radiation pattern $A_{E,V}(\vartheta)$, which can be represented as

$$\begin{aligned} A_E(\vartheta, \phi) &= G_{E,\max} - \min\{-[A_{E,V}(\vartheta) + A_{E,H}(\phi)], A_m\} \\ A_{E,V}(\vartheta) &= -\min\left\{12\left(\frac{\vartheta - 90^\circ}{\vartheta_{3\text{dB}}}\right)^2, SLA_v\right\} \\ A_{E,H}(\phi) &= -\min\left\{12\left(\frac{\phi}{\phi_{3\text{dB}}}\right)^2, A_m\right\} \end{aligned} \tag{1}$$

where ϑ and ϕ denote the elevation angle and azimuth angle from UAV toward a cell of the BS, respectively, $\vartheta_{3\text{dB}} = 65^\circ$ and $\phi_{3\text{dB}} = 65^\circ$ are the vertical and horizontal 3dB beamwidths, respectively, $G_E^{\max} = 8$ dBi is the maximum directional gain of antenna element, $A_m = 30$ dB is the front-back ratio, and $SLA_v = 30$ dB is the side-lobe level limit. Subsequently, according to Table 5.4.4.2-2 in [29], the array radiation pattern

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can be obtained by

$$G_A(\vartheta,\phi) = A_E(\vartheta,\phi) + 10\log_{10}\left[1+\chi\left(\left|\sum_{l=1}^{L} w_l \cdot v_l\right|^2 - 1\right)\right] \quad (2)$$

where $\chi \in [0,1]$ represents the correlation coefficient, $L=8$ is the number of array antenna elements, $v_l = e^{(\frac{2\pi}{\lambda} j(l-1)d_V \cos(\vartheta))}$ and $w_l = \frac{1}{\sqrt{L}} e^{(\frac{2\pi}{\lambda} j(l-1)d_V \sin(\vartheta_{etilt}))}$ are the steering vector components and weighting factor, respectively.

As the BS has three cells, (ϑ,ϕ) is distinct when the UAV is associated with different cells. The direction towards to the cells $b \in \{1,2,3\}$ at time t is denoted as $(\vartheta_b, \phi_b; \mathbf{q}(t))$. The UAV is associated with the cell that has the highest antenna gain, which can be expressed as

$$G_{BS}(t) = \max_{b\in\{1,2,3\}} G_A(\vartheta_b, \phi_b; \mathbf{q}(t)) \quad (3)$$

The system environment is assumed as a rural environment with almost no obstructions or scattering, where the communication channels between UAVs and ground nodes are dominated by the LoS links. In practice, a reasonable approximation of the communication channel gain can be obtained by a free-space path loss model [30], [31]. Therefore, the communication channel gain between UAV and BS can be expressed as

$$h_{com}(t) = \frac{G_t G_{BS}(t)\lambda^2}{(4\pi)^2 d_{BU}^2(t)}, \forall t \in [0,T] \quad (4)$$

where $d_{BU}(t) = \|\mathbf{q}(t) - \mathbf{g}\|$ denotes the distance between the UAV and the BS at time t, and G_t denotes the UAV transmitting antenna gain.

The communication channel gain between the UAV and the mth IoT node can be expressed as

$$\hat{h}_{com}^m(t) = \frac{G_v G_r \lambda^2}{(4\pi)^2 d_m^2(t)}, \forall m \in M, \forall t \in [0,T] \quad (5)$$

where $d_m(t) = \|\mathbf{q}(t) - \mathbf{g}_m\|$ represents the distance between the UAV and the mth node at time t, G_v and G_r denote the node transmitting antenna gain and UAV receiving antenna gain, respectively.

C. ISAC Power and Sensing Order

Due to that the positions of the IoT nodes are unknown to the UAV, the order of sensing nodes is determined based on the power strength of the IoT status signals received by the UAV. The status signals from different IoT nodes are transmitted by frequency division multiple access (FDMA). That is the total system bandwidth B is divided equally into M subchannel, which are assigned to M IoT nodes, respectively. Hence, the UAV can identify the signals from different IoT nodes by searching the full frequency band. The additive noise is not considered in the status signals, which is small relative to the signal. The signal power strength received by the UAV from the mth IoT node at time t can be expressed as

$$\hat{P}_{com}^m(t) = P_0 \hat{h}_{com}^m(t), \forall m \in \mathcal{M}, \forall t \in [0,T] \quad (6)$$

where P_0 is the transmit power for each node.

The UAV radar sensing analyzes the directly reflected echo signal from the target and extracts the relevant information. According to [32], the average power gain of the radar sensing link to the mth IoT node can be expressed as

$$h_{rad}^m(t) = \frac{G_t G_{rad} \sigma_{RCS}^m \lambda^2}{(4\pi)^3 d_m^4(t)} \quad (7)$$

where σ_{RCS}^m denotes the radar cross section (RCS) of the mth target, and G_{rad} denotes the radar receiving antenna gain of the UAV.

To perform the ISAC function, we assume that the transmit power of the UAV is split into two power streams, one for communicating with the BS and the other for sensing the IoT nodes. Without loss of generality, we define the maximum sensing coverage angle of the UAV as θ. To save the energy consumption of the UAV, the radar sensing signal should be transmitted when the mth target node is within the UAV radar coverage, which can be denoted as

$$d_m(t) \le d_{rad} \quad (8)$$

where $d_{rad} = \frac{H}{\cos\theta}$. As the location of IoT nodes is unknown to the UAV, the distance $d_m(t)$ between the UAV and the mth node is not available. According to (5) and (6), the distance between UAV and node can be estimated by the strength of the status signal received by the UAV. Hence, to determine whether a node is within the radar sensing coverage, the receive power threshold P_{th} can be expressed as

$$P_{th} = P_0 \frac{G_v G_r \lambda^2}{(4\pi)^2 d_{rad}^2} \quad (9)$$

Assuming the total transmit power of the UAV as P, the transmit power for UAV radar sensing at time t can be expressed as

$$P_{\text{rad}}(t) = \begin{cases} 0, & \hat{P}_{com}^m(t) < P_{th} \\ \kappa P, & \hat{P}_{com}^m(t) \ge P_{th} \end{cases} \quad (10)$$

where κ is the power allocation factor. Hence, the transmit power for communications can be given as $P_{\text{com}}(t) = P - P_{\text{rad}}(t)$. When the UAV is sensing IoT nodes, it will simultaneously maintain communication with the base station.

Obviously, sensing the IoT node closest to the UAV is the shortest trajectory to complete the radar sensing. For achieving the shortest flight time, the m^*th target node for radar sensing is determined by

$$m^* = \arg\max_{m\in\mathcal{M}} \hat{P}_{com}^m(t) \quad (11)$$

We denote the number of the nodes sensed by the UAV at time t as $J(t)$. If there is no sensed IoT node, the UAV will maintain the same value of $j(n)$ as the previous time step. When the UAV completes sensing m^*th IoT node, $J(t) = J(t)+1$ and then remove m^* from $\mathcal{M}$.

III. UAV Trajectory Optimization Formulation

In this section, to evaluate the sensing performance of the UAV-ISAC assisted IoT system, we discuss a new performance

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metric for radar sensing, namely radar estimation rate. Subsequently, we present the UAV trajectory optimization problem to achieve both better sensing performance and shorter flight time.

A. System Performance Evaluation Metrics

In radar systems, radar estimation rate has been proposed to measure the radar sensing performance from the perspective of information theory [33]. The radar estimation rate is defined as the MI between the radar and the target. According to information theory, the larger the MI, the greater the reduction in prior uncertainty after measurement, resulting in more information gained about the target. Therefore, the radar estimation rate is an important measure metric for the radar system, which can be maximized to improve the radar sensing ability [34], [35]. To evaluate the performance of the UAV-ISAC, we use radar estimation rate and information rate as the performance metrics for the radar and communication functions, respectively.

At time t, considering the interference between sensing and communication, the UAV radar signal-to-interference-plus-noise ratio (SINR) for the mth node and the communication SINR for the BS are respectively given by

$$\gamma_{rad}^{m}(t) = \frac{P_{rad}(t)h_{rad}^{m}(t)}{N_0 B + P_{com}(t)h_{rad}^{m}(t)} \tag{12}$$

$$\gamma_{com}(t) = \frac{P_{com}(t)h_{com}(t)}{N_0 B + P_{rad}(t)h_{com}(t)} \tag{13}$$

where N_0 is the power spectrum density of the Additive White Gaussian Noise (AWGN).

Based on Shannon's theorem [36], the information rate of the BS at time t can be expressed as

$$c_{com}(t) = B\log_2\left(1 + \gamma_{com}(t)\right) \tag{14}$$

According to [33], the radar estimation rate for sensing the mth node at time t can be represented as

$$c_{rad}^{m}(t) = B\log_2\left(1 + \gamma_{rad}^{m}(t)\right) \tag{15}$$

Therefore, the total information rate for the BS and the total radar estimation rate for the mth node can be respectively expressed as

$$C_{com} = \int_{t=0}^{T} c_{com}(t)dt \tag{16}$$

$$C_{rad}^{m} = \int_{t=0}^{T} c_{rad}^{m}(t)dt \tag{17}$$

And the system radar estimation rate for all the nodes is defined as

$$C_{rad} = \sum_{m=1}^{M} C_{rad}^{m} \tag{18}$$

In addition, to obtain the accurate sensing information for each node, the total radar estimation rate of each node must be greater than a threshold, which is expressed as

$$C_{rad}^{m} \geq C_{th}, \forall m \in \mathcal{M} \tag{19}$$

where C_{th} represents the minimum radar estimation rate required for accurate target detection. To ensure that the UAV can forward all the sensing information to the BS, the information rate for the BS should be greater than the system radar estimation rate, which is denoted as

$$C_{com} \geq C_{rad} \tag{20}$$

B. Problem Formulation

In the UAV-ISAC assisted IoT system, the mission of the UAV is to complete the sensing of M IoT nodes during the flight. Considering the main task and the limited energy of the UAV, the UAV trajectory is designed to maximize the radar estimation rate C_{rad} while minimizing the UAV flight time T. Since the UAV can adjust its trajectory more flexibly to obtain larger C_{rad} with the increase of flight time, there is an inherent trade-off between the flight time and the system radar estimation rate. To characterize this tradeoff, we introduce a weight factor $\alpha \leq 0$ to balance it. Thus, the optimization problem is formulated as follows

$$P(1): \max_{\mathbf{q}(t)} \quad C_{rad} + \alpha T \tag{21a}$$

$$\text{s.t.} \quad \|\dot{\mathbf{q}}(t)\| = V, \ \forall t \in [0, T] \tag{21b}$$

$$\mathbf{q}(t) \in \mathcal{F}, \ \forall t \in [0, T] \tag{21c}$$

$$\mathbf{q}(0) = \mathbf{q}_I, \ \mathbf{q}(T) = \mathbf{q}_I \tag{21d}$$

$$C_{rad}^{m} \geq C_{th}, \forall m \in \mathcal{M} \tag{21e}$$

$$C_{com} \geq C_{rad} \tag{21f}$$

$$J(T) = M \tag{21g}$$

where the constraint (21b) indicates that the UAV has a fixed velocity V, the constraint (21c) ensures that the UAV is deployed only in the area $\mathcal{F}$, the constraint (21d) indicates that the UAV returns to the starting point to finish the flight, the constraint (21e) clarifies the minimum sensing requirements for each IoT node, the constraint (21f) ensures that all the sensing information can be sent to the BS, and the constraint (21g) indicates that the UAV needs to sense all the M nodes during the flight period.

Directly solving the optimization problem P(1) will encounter the following problems. The locations of the IoT nodes are unknown, thus, it is impossible to derive a closed-form expression for the radar estimation rate. Due to the unknown IoT node positions and the non-convex nature of the optimization problem proposed, traditional convex optimization techniques struggle to find a solution. The superiority of DRL lies in its ability to learn through the interaction between the agent and the environment, without requiring complete information about the environment or assuming convexity of the problem. Therefore, in the face of challenges such as a lack of substantial prior system knowledge and the non-convex nature of the optimization problem, adopting DRL technology for the optimization problem presented in this paper is highly necessary. In conclusion, we propose a DRL-based UAV trajectory optimization algorithm to solve the UAV trajectory optimization problem.

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IV. Deep Reinforcement Learning Algorithm Design

In this section, we provide a brief introduction to the DRL method, and then propose an algorithm bsaed on multi-step learning Dueling DDQN to solve the UAV trajectory optimization problem.

A. Deep Reinforcement Learning

RL focuses on how the agent takes the actions in the environment to maximize the cumulative reward. A fully observable MDP can be represented by the 4-element tuple combination $< \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R} >$, where $s \in \mathcal{S}$ represents the state of agent, $a \in \mathcal{A}$ represents the actions that an agent can perform, $\mathcal{P}$ represents the state transition probability space, and $\mathcal{R}$ represents the reward. $P(S_{n+1}|S_n, A_n) \in \mathcal{P}$ denotes the probability of the agent's transition from the current state $S_n \in \mathcal{S}$ to the next state $S_{n+1} \in \mathcal{S}$ after taking action $A_n \in \mathcal{A}$. $R_n \in \mathcal{R}$ is the immediate reward for the agent taking action A_n in state S_n. The reward function determines the learning goal of the agent.

The agent takes actions based on the strategy π. The RL method aims to find an optimal strategy π^* that enables the agent to maximize its cumulative rewards or achieve its goals in the most effective way when interacting with the environment. The merits of strategies are typically compared using the corresponding value functions. For example, a commonly used value function is the action-value function $Q_\pi(S_n, A_n)$ defined as follows

$$Q_\pi\left(S_n, A_n\right) = \mathbb{E}_\pi\left[G_n | S_n, A_n\right] \tag{22}$$

where $G_n = \sum_{i=0}^{\infty} \eta^i R_{n+i}$ is the long-term cumulative reward obtained by the agent after taking action A_n following strategy π starting from the current state S_n, $0 \leq \eta \leq 1$ is the discount factor, and $\mathbb{E}_\pi$ denotes the expectation operation on the cumulative reward.

The optimal strategy π^* can be determined by finding the optimal action-value function $Q^*(S_n, A_n)$, which is defined as

$$Q^*(S_n, A_n) = \max_{\pi} Q_\pi(S_n, A_n) \tag{23}$$

During the agent's learning process, the update of action-value function $Q^*(S_n, A_n)$ is denoted as

$$Q^*(S_n, A_n) = R_n + \eta \sum_{s' \in \mathcal{S}} P(s'|S_n, A_n) \max_{a' \in \mathcal{A}} Q^*(s', a') \tag{24}$$

where s' represents all possible next states, and a' represents all possible actions to be taken in the next state.

Q-learning algorithm requires maintaining a huge Q-table under larger or continuous state spaces, which makes the lookup inefficiently. In order to apply the RL algorithm practically to this situation, function approximation can be applied to solve such problems. The function approximation method fits the true action-value function by $\hat{Q}(S_n, A_n; \omega) \approx Q(S_n, A_n)$. Artificial neural network (ANN) technology possesses the function approximation capability to meet these requirements. The powerful DRL framework combines RL with functional approximation using ANN. One popular approach that combines Q-learning algorithms with ANN is the deep Q-learning network. The parameter ω of $\hat{Q}(S_n, A_n; \omega)$ corresponds to the weight coefficients and biases in ANN, and the training process of the $\hat{Q}(S_n, A_n; \omega)$ function involves updating the values of ω. Standard ANNs are typically trained using supervised learning techniques, but DRL is not well-suited for this approach. To address this issue, bootstrapping is employed in DRL, where the network parameters are updated using a minimization loss, which is given by

$$f_{loss} = \left(R_n + \eta \max_{a \in A} \hat{Q}(S_{n+1}, a; \omega) - \hat{Q}(S_n, A_n; \omega)\right)^2 \tag{25}$$

The loss function (25) often leads to the issue of overestimation. Some studies [37], [38], [39] have proposed the use of target networks $\hat{Q}(S_n, A_n; \omega^-)$ for action evaluation in order to address the issue of overestimation. As a result, the loss function is modified as

$$f_{loss} = \left(R_n + \eta \max_{a \in A} \hat{Q}(S_{n+1}, a; \omega^-) - \hat{Q}(S_n, A_n; \omega)\right)^2 \tag{26}$$

The target network $\hat{Q}(S_n, A_n; \omega^-)$, which is updated by the main network $\hat{Q}(S_n, A_n; \omega)$, is updated less frequently compared to $\hat{Q}(S_n, A_n; \omega)$. For example, ω^- can be set to ω after every Z updates of the weights ω in the main network $\hat{Q}(S_n, A_n; \omega)$, and kept unchanged for the next Z updates of ω.

B. Algorithm Design

Since MDP is typically defined in discrete time steps, we discretize the flight time T into N time steps with a time interval of Δ_t. Then, the information rate $c_{com}(t)$ and the radar estimation rate $c_{rad}^m(t)$ can be respectively represented in discrete time as $c_{com}(n)$ and $c_{rad}^m(n)$ for $n \in \{1, 2, \ldots, N\}$. As a result, the information rate C_{com} and the total radar estimation rate for the mth node C_{rad}^m can be approximately rewritten as

$$\tilde{C}_{com} \approx \Delta_t \sum_{n=1}^{N} c_{com}(n) \tag{27}$$

$$\tilde{C}_{rad}^m \approx \Delta_t \sum_{n=1}^{N} c_{rad}^m(n) \tag{28}$$

Then, the optimization problem $P(1)$ can be rewritten as

$$P(2): \max_{\mathbf{q}(n)} \; C_{rad} + \alpha N \tag{29a}$$

$$\text{s.t.} \; \|\dot{\mathbf{q}}(n)\| = V, \; \forall n \in [0, N] \tag{29b}$$

$$\mathbf{q}(n) \in \mathcal{F}, \; \forall n \in [0, N] \tag{29c}$$

$$\mathbf{q}(0) = \mathbf{q}_I, \; \mathbf{q}(N) = \mathbf{q}_I \tag{29d}$$

$$\tilde{C}_{rad}^m \geq C_{th}, \; \forall m \in \mathcal{M} \tag{29e}$$

$$\tilde{C}_{com} \geq C_{rad} \tag{29f}$$

$$J(N) = M \tag{29g}$$

Subsequently, we can obtain a 4-element tuple combination $< \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R} >$ as specified below.

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- States space $\mathcal{S}$: under different task phases of sensing the targets or returning to the starting location, the UAV obtains different reward values. We set the state by combining the UAV position with the sensed target during the sensing phase, i.e., $S_n = [\mathbf{q}(n), j(n)], j(n) \in 1, 2, \ldots, M$, where $j(n)$ represents the sensing target. When the UAV returns to the starting location after completing sensing all nodes, the state is represented as $S_n = [\mathbf{q}(n), M+1]$. Therefore, the state space $\mathcal{S}$ is denoted as

$$\mathcal{S} = \{S_n \,|[\,\mathbf{q}(n), j(n)], j(n) \in \{1, 2, \ldots, M+1\}\} \tag{30}$$

  When the UAV flies away from the area $\mathcal{F}$ or the information rate is less than the system radar estimation rate (constraints (29b) and (29e) are not satisfied), the state is marked as $S^{(1)}$. During the sensing phase, the UAV selects the m^*th IoT node as the target according to (11). When the UAV completes the sensing for the m^*th node (constraint (29d) is satisfied), the state is marked as $S^{(2)}$.
- Actions space $\mathcal{A}$: action $A_n \in \mathcal{A}$ represents the moving direction of the UAV at time n. We set $K = 8$ moving directions of the UAV, which are separated by angle of 45° each. Therefore, the action space is denoted as

$$\mathcal{A} = \left\{A_n | a^{(k)}, k \in \{1, 2, \ldots, K\}\right\} \tag{31}$$

- Transition probabilities space $\mathcal{P}$: $\mathcal{P}$ denotes the probability of UAV's state transition, which is related to the current state S_n and the selected action A_n.
- Rewards space $\mathcal{R}$: $R_n \in \mathcal{R}$ is the immediate reward for the UAV taking action A_n in state S_n. When the state $S_n = S^{(1)}$, the UAV will receive a penalty $-R_0$. When the state $S_n = S^{(2)}$, it will receive a reward $R_1 = \rho C_{th}$ where ρ is the weight factor. When the UAV reaches the starting location $\mathbf{q}_I$ after completing the sensing of M nodes, i.e., $S_n = [\mathbf{q}_I, M+1]$, it will receive a reward R_2. UAV receives a penalty $\beta c^m_{rad}(n) - R_{step}$ for each time step, where β is the weight factor and R_{step} is the penalty. The learning objective of the algorithm is to find the policy that maximizes the cumulative reward. Therefore, the penalty $\beta c^m_{rad}(n) - R_{step}$ at each step should be negative, encouraging the UAV to minimize flight time as much as possible. Additionally, in order to maximize the system's radar estimation rate, the weight of radar estimation rate β is set to a relatively large value while ensuring that the penalty at each step remains negative. Combining the objective function (29a) and the penalty at each step, the weight factor α of the objective function (29a) is set as $\alpha = -\frac{R_{step}\Delta_t}{\beta}$.
  Accordingly, the reward function is designed as follows

$$R_n = \begin{cases} -R_0, & \text{if } S_n = S^{(1)}; \\ R_1, & \text{if } S_n = S^{(2)}; \\ R_2, & \text{if } S_n = [\mathbf{q}_I, M+1]; \\ \beta c^m_{rad}(n) - R_{step}, & \text{else.} \end{cases} \tag{32}$$

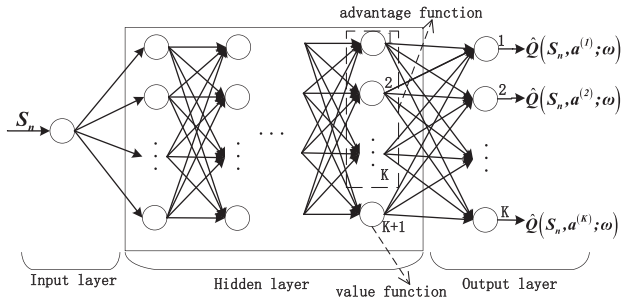


Fig. 2. Dueling Network Architecture.

After defining the sets $\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}$, we can apply DQN algorithm to solve the optimization problem $P(2)$. The traditional architecture of the ANN used in DQN algorithm typically consists of three parts: an input layer, one or more hidden layers, and an output layer. The input layer takes the state representation S_n as input, and the hidden layers perform computations to learn meaningful features and representations of the state-action pairs. The output layer produces the estimated action values for each possible action $a^{(k)}$. The proposed algorithm adopts the Dueling network architecture, which is illustrated in Fig. 2. It can be seen that Dueling network architecture includes two sub-networks between the hidden layer and the output layer. The first sub-network is solely related to the state S_n and represents the value function, while the second sub-network is related to both the state S_n and the action $a^{(k)}$, and represents the advantage function. The output of the Dueling network is obtained by combining the output of the value function and the advantage function. The use of the Dueling network architecture can lead to a more stable approximation error compared to traditional DQN, especially when the value gap of $Q(S_n, A_n)$ between different actions for the same state is small.

The update of $Q(S_n, A_n)$ always based on the one-step TD learning, but using information from only one step ahead may result in poor predictive performance for making decisions. To address this limitation, we adopt multi-step TD learning method with the number of steps denoted as N_{step}. Multi-step TD learning incorporates information from multiple steps ahead to update the action values, allowing for a more accurate estimation of the action values. Then, the N_{step}-step reward is expressed as

$$R_{n:n+N_{step}} = \sum_{i=0}^{N_{step}-1} \eta^i R_{n+i+1} \tag{33}$$

The loss function of the multi-step learning DDQN algorithm can be expressed as

$$\begin{aligned} f_{loss} = (&R_{n:n+N_{step}} \\ &+ \eta^{N_{step}} \hat{Q}(S_{n+N_{step}}, a^*; \omega^-) - \hat{Q}(S_n, A_n; \omega))^2 \end{aligned} \tag{34}$$

where $a^* = \arg\max_{a^{(k)} \in \mathcal{A}} \hat{Q}(S_{n+N_{step}}, a^{(k)}; \omega)$.

In the fitting process of function $Q(S_n, A_n)$, the issue of sample correlation usually arises due to the high correlation

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Algorithm 1: UAV Trajectory Optimization Algorithm Based on Multi-Step Learning Dueling DDQN

Initialize: the maximum number of episodes N_{epi}, maximum number of steps per episode N_{max}, initial exploration ε_0, decaying rate ξ, set $\varepsilon \leftarrow \varepsilon_0$;
Initialize: the reply buffer $\mathbf{E}$ with capacity D, the main Dueling network with parameters ω, the target Dueling network with parameters $\omega^- = \omega$;
1: **for** episode: $n_{epi} = 1, 2, \ldots, N_{epi}$ **do**
2: initialize the sliding window queue $\mathbf{W}$ with capacity N_{step};
3: initialize UAV state S_0;
4: **repeat**
5: select action $A_n \in \mathcal{A}$ based on ε-greedy strategy, where $\varepsilon \leftarrow \varepsilon \xi^{n_{epi}}$;

$$A_n = \begin{cases} \underset{a^{(k)} \in \mathcal{A}}{\text{argmax}}\, \hat{Q}(S_n, a^{(k)}; \omega), \text{ prob. of } (1-\varepsilon); \\ \text{random action, prob. of } \varepsilon \end{cases}$$

6: perform action A_n and observe the next state S_{n+1}, obtain reward R_n according to (32);
7: store (S_n, A_n, R_n, S_{n+1}) in $\mathbf{W}$;
8: **if** $n \geq N_{step}$ **then**
9: calculate the cumulative reward of N_{step} steps according to (33), then store $(S_{n-N_{step}}, A_{n-N_{step}}, R_{(n-N_{step}):n}, S_n)$ in $\mathbf{E}$;
10: **end if**
11: randomly sample mini-batch of N_{step} step transition $(S_i, A_i, R_{i:i+N_{step}}, S_{i+N_{step}})$ from $\mathbf{E}$; calculate

$$y_i = \begin{cases} R_{i:i+N_{step}} - R_0, \ if\ S_{i+N_{step}} = S^{(1)}; \\ R_{i:i+N_{step}} + R_1, \ if\ S_{i+N_{step}} = S^{(2)}; \\ R_{i:i+N_{step}} + R_2, if\ S_{i+N_{step}} = [\mathbf{q}_I, M+1]; \\ R_{i:i+N_{step}} + \eta^{N_{step}} \hat{Q}(S_{i+N_{step}}, a^*; \omega^-), \\ \qquad\qquad \text{otherwise.} \end{cases}$$

12: Execute a gradient descent step on $(y_i - \hat{Q}(S_i, A_i; \omega))^2$ in regard to parameters ω;
13: Update $n_{epi} \leftarrow n_{epi} + 1$;
14: **until** $S_n = [\mathbf{q}_I, M+1]$ or $S_n = S^{(1)}$ or $n = N_{max}$.
15: After every Z_{epi} eposodes, update the target network parameter with $\omega^- = \omega$;
16: **end for**

between consecutive samples generated in a continuous time series. This can lead to instability and slow convergence of the algorithm. One common solution to address sample correlation is to use experience replay, which stores the samples $(S_n, A_n, R_{n:(n+N_{step})}, S_{n+N_{step}})$ in a replay buffer $\mathbf{E}$ and gets these samples from buffer $\mathbf{E}$ during the network training process. Randomly sampling a batch from $\mathbf{E}$ helps to break the sequential correlation between consecutive samples.

Accordingly, we adopt multi-step learning Dueling DDQN technology for UAV trajectory optimization. Algorithm 1 outlines the steps of the proposed algorithm. Step 2 to 13 makes up an episode. In Step 2, a sliding window queue W with capacity N_{step} is initialized to calculate the cumulative reward for N_{step} steps. In Step 3, the state S_0 is randomly initialized for training. In Step 5, actions are selected based on the ε-greedy strategy, where the UAV randomly chooses action A_n with probability ε, or selects the action corresponding to the highest value of the action-value function with the probability of $1-\varepsilon$. The value of ε decays with an increase in episodes. In Step 6, the reward is calculated based on (33), and the transfer sample (S_n, A_n, R_n, S_{n+1}) are stored in the sliding window queue $\mathbf{W}$. Steps 8, 9 and 10 calculate the N_{step}-step cumulative reward $R_{n:n+N_{step}}$ and store the transfer samples $(S_{n-N_{step}}, A_{n-N_{step}}, R_{(n-N_{step}):n}, S_n)$ in the buffer $\mathbf{E}$. DDQN learning is applied in Step 11. Step 12 executes the gradient descent step according to (34). Step 14 denotes that an episode will terminate when $S_n = [\mathbf{q}_I, M+1]$ or $S_n = S^{(1)}$ or $n = N_{max}$. $n = N_{max}$ indicates that the UAV flight time has reached the maximum flight time N_{max}. Step 15 updates the target Dueling network parameters by $\omega^- = \omega$ after every Z_{epi} episodes.

The time complexity of Algorithm 1 is calculated as follows. The time complexity of state representation is $O(1)$. The time complexity of forward propagation and back propagation is $O(L \times M)$, where L is the number of network layers and M is the number of neurons per layer. Each update requires randomly selecting N_{step} samples from the replay buffer $\mathbf{E}$. This process is $O(N_{step})$. Therefore, the time complexity of Algorithm 1 is $O(N_{epi} \times (L \times M + N_{step}))$. The space complexity of Algorithm 1 is mainly determined by the Q-value network and the replay buffer $\mathbf{E}$. The space complexity of the Q-value network is mainly determined by the number of network layers L and the number of neurons in each layer M, which is simplified to $O(L \times M^2)$. The space complexity of the replay buffer $\mathbf{E}$ is $O(D)$, depending on the capacity D. Therefore, the space complexity of Algorithm 1 is $O(L \times M^2 + D)$.

The convergence of Algorithm 1 is theoretically guaranteed [37]. However, in the early stages of training, the effective trajectory for UAV exploration to complete the sensing of M IoT nodes can potentially be time-consuming. Therefore, we propose an initialization of $\hat{Q}(S_n, A_n; \omega)$. Specifically, the UAV is encouraged to approach the target based on the received signal power strength from the IoT nodes during the sensing phase, while the UAV is encouraged to return to $\mathbf{q}_I$ based on the distance between UAV and starting location during the returning phase. The simulation results in the next section will verify that the proposed algorithm strikes a balance between maximizing the system radar estimation rate and minimizing UAV flight time.

V. Simulations and Discussions

In this section, we conduct simulation experiments to validate the superiority of the proposed algorithm. We first demonstrate the convergence of the proposed algorithm through simulation results at different parameter values of C_{th}. Furthermore, we compared the proposed algorithm with benchmark methods.

A. Simulation Settings

As shown in Fig. 3, we consider an area with a plane size of 1.5 km $\times$ 1.5 km. It has one BS with a location of $\mathbf{g} = [200, 800, 25]$ m, and $M = 8$ IoT nodes are randomly distributed

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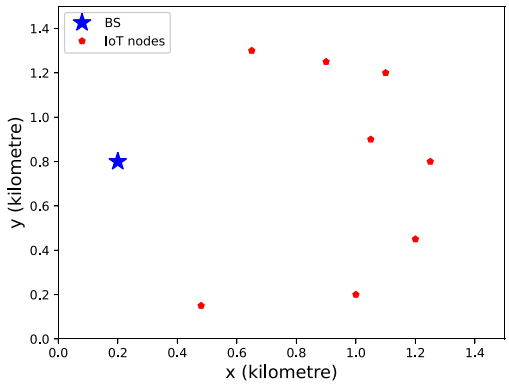


Fig. 3. The BS location and IoT nodes location in 2D views.

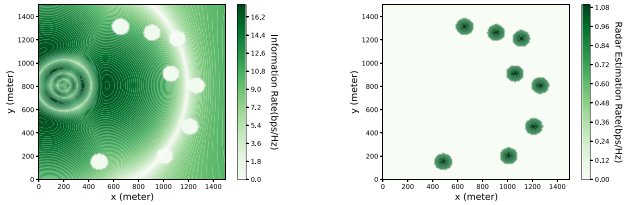
TABLE II
MAIN SIMULATION PARAMETERS

Parameters	Values
bandwidth B [28]	100 MHz
signal wavelength λ	0.15 m
node transmit power P_0	2 W
UAV transmit power P	10 W
maximum sensing angle of UAV θ	30°
RCS of the target σ_{RCS}^m	1 m^2
UAV speed V	20 m/s
time interval Δ_t	0.5 s
noise power spectrum density N_0 [29]	–174 dBm/Hz
array correlation coefficient χ	1
maximum number of episode N_{epi}	3000
maximum number of steps per episode N_{max}	1500
initial exploration rate ε_0 [37]	0.5
exploration decaying rate ξ	0.997
discount factor η	1
initial exploration rate ε_0 [37]	30
capacity for replay buffer D	5×10^6
update frequency for target network Z_{epi} [37]	5
penalty $-R_0$	–80000
reach destination reward R_2	15000
penalty per step $-R_{step}$	–41
weight factor ρ	4×10^{-6}
weight factor β	1.6×10^{-7}

at a distance from the BS. The flight altitude of the UAV is fixed at $H = 100$ m, and the starting position of the UAV is $\mathbf{q}_I = [200, 800, 100]$ m [37]. Additionally, the power allocation factor κ of the UAV is set to 0.6. The antenna gains G_v, G_t, G_r, and G_{rad} are constants with the value of 2.15 dBi [28] [29]. The values of the other parameters used for the simulation experiments are summarized in Table II.

The Dueling network used in the proposed algorithm is a fully connected ANN with five hidden layers. The first four hidden layers have 512, 256, 128, and 128 neurons, respectively. The last layer among the hidden layers is the Dueling layer with $K + 1$ neurons. The Rectified Linear Unit (ReLU) activation function is used for the first four hidden layers, and the linear activation function is used for the Dueling layer. The ANN is trained with the Adam optimizer.

Based on the simulation settings above, we can obtain the real communication information rate map of the BS and the real radar estimation rate map of the UAV in the area $\mathcal{F}$, as shown in Fig. 4(a) and (b), respectively. It's worth noting that the UAV is not able to directly obtain the location information of the nodes.



(a) (b)

Fig. 4. True information rate of BS and radar estimation rate of UAV in area $\mathcal{F}$.

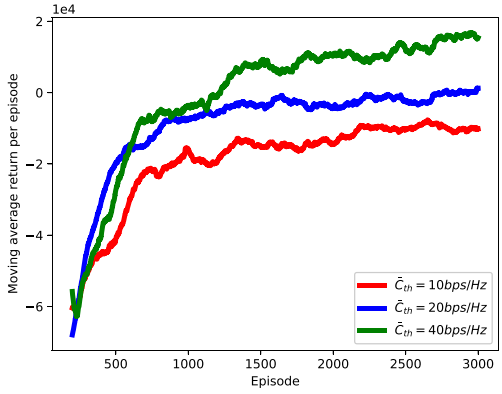


Fig. 5. Moving average returns under different values of C_{th}.

Fig. 4(a) shows the presence of zero-value regions, which are caused by the existence of sidelobe zero points in the antenna.

B. Proposed Algorithm Performance Analysis

To demonstrate the convergence of the proposed algorithm, we conducted simulation experiments in three scenarios where the values of C_{th} were set differently. Specifically, the values of C_{th} are set to 5 bps/Hz, 10 bps/Hz, and 20 bps/Hz, respectively. The simulation results of the proposed algorithm under these three scenarios are presented as follows.

Fig. 5 displays the moving average returns of the proposed algorithm under different values of C_{th}. As evident from the figure, the moving average returns of the algorithm increase with the number of episodes, eventually stabilizing. Additionally, the moving average returns of the algorithm stabilize at approximately -11600, -4750, and 8485, respectively, for different values of C_{th}. Notably, the system radar estimation rate, which is one of the reward indicators according to (32), increases with different values of C_{th}, leading to higher moving average returns. It is observed that the algorithm under $C_{th} = 20$ bps/Hz converge to a stable value after 2000 episodes, whereas the algorithm stabilizes after 1500 episodes for other values of C_{th}. This is attributed to the fact that the algorithm under $C_{th} = 20$ bps/Hz requires more flight time for exploration, as corroborated by the analysis of UAV flight time in the subsequent sections. This indicates that the proposed algorithm is capable of learning a convergent and stable strategy under different values of C_{th}.

The UAV trajectory design aims to maximize the radar estimation rate while minimizing UAV flight time, subject to the

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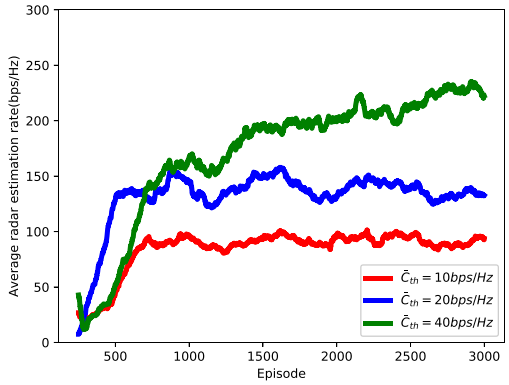


Fig. 6. Average radar estimation rate under different values of C_{th}.

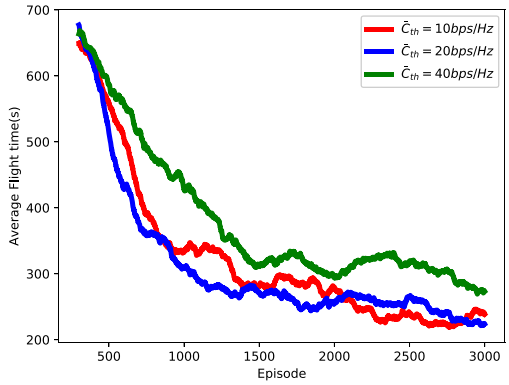


Fig. 7. UAV flight time under different values of C_{th}.

constraints defined in (29b)–(29g). However, during the training process, it is possible that the UAV may not complete its sensing mission and end its flight, resulting in that the UAV is in state $S^{(1)}$ or continues to fly until it reaches the maximum flight time N_{max}. In such cases, the UAV trajectory do not satisfy the constraints of the optimization problem $P(2)$. To evaluate the performance of the algorithm in achieving the optimization objective while satisfying the constraints, we set the flight time and the system radar estimation rate to N_{max} and 0 bps/Hz, respectively, when the UAV trajectory fails to satisfy the constraints. With this setup, we can now discuss the UAV flight time and the system radar estimation rate of the algorithm under different values of C_{th}.

Fig. 6 illustrates the average system radar estimation rate of the proposed algorithm under different values of C_{th}. It is evident that as C_{th} increases, the UAV performs more steps for sensing IoT nodes, resulting in a higher radar estimation rate. The values of C_{rad} in Fig. 6 exhibit a decreasing trend, which can be attributed to the initialization of $\hat{Q}(S_n, A_n; \omega)$ guiding the UAV to complete the task. As the number of training episodes increases, the system radar estimation rate also increases, indicating that the UAV is able to complete the task of sensing multiple IoT nodes through the training process. Furthermore, it can be observed that as the number of episodes increases, the radar estimation rate also increases and eventually converges to approximately 48 bps/Hz, 63 bps/Hz, and 100 bps/Hz under different values of C_{th}, respectively.

We have studied the average flight time of the UAV in the proposed algorithm under different values of C_{th}, as depicted in Fig. 7. It can be observed that as the number of episodes increases, the UAV flight time becomes shorter and eventually converges. Furthermore, it is evident that a larger C_{th} results in a longer flight time for the UAV. This is consistent with the fact that a larger C_{th} value leads to more steps for the UAV to sense the IoT nodes. By combining the findings from Figs. 6 and 7, it can be concluded that the proposed algorithm achieves joint optimization of the system radar estimation rate and flight time, while adhering to the constraint of returning to the starting point after completing the sensing of all the IoT nodes.

Fig. 8 displays the simulated UAV trajectory obtained from the algorithm with different values of C_{th}. In order to provide an intuitive view of the UAV flight trajectory, we present the plane

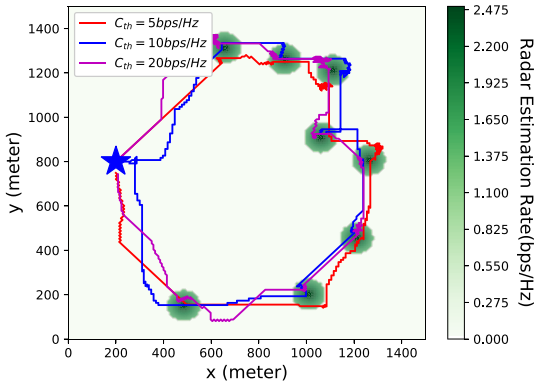


Fig. 8. UAV trajectory under different threshold values of C_{th}.

trajectory from an overhead perspective. To better visualize the UAV sensing for IoT nodes, we utilize the real radar estimated rate map as the background of the UAV plane trajectory. From Fig. 8, it can be observed that the UAV is able to return to the starting location after successfully completing the sensing of M IoT nodes, regardless of the value of C_{th}. Furthermore, it is evident that the UAV trajectory, trained using the proposed algorithm, covers a larger sensing area when the value of C_{th} is larger.

C. Performance Comparison With Benchmark Methods

In this section, we aim to verify the superiority of the proposed algorithm by comparing its performance with benchmark methods. Firstly, we present the RL-based benchmark methods below. It is important to note that all algorithms utilize the same parameters unless stated otherwise.

- **DDQN**: the algorithm adopts the DDQN to approximate the true action-value function.
- **Greedy**: the UAV only takes the action that corresponds to the largest value of action-value function in the current state. Besides, the network structure is consistent with the proposed algorithm.
- **Position-based**: the algorithm sets the state only based on the position of the UAV, i.e., $S_n = \mathbf{q}(n)$.
- **Random**: the UAV randomly selects an action from $\mathcal{A}$ at each time slot.

Furthermore, the proposed algorithm is also compared individually with the direct flight method without RL. Specifically,

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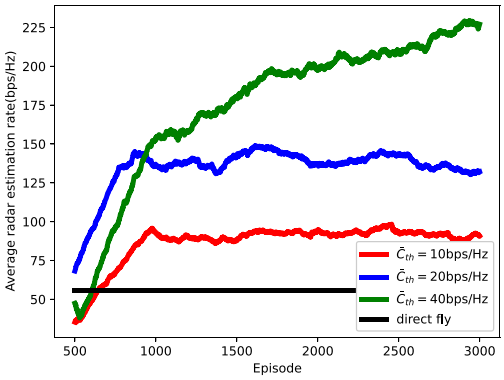


Fig. 9. Average radar estimation rate of the proposed algorithm and the direct flight method.

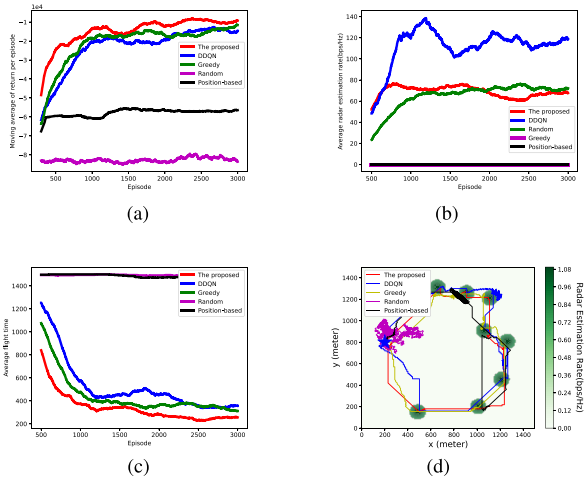


Fig. 10. Performance comparison versus different algorithms under $C_{th} = 5$ bps/Hz. (a) Moving average returns. (b) average radar estimation rate. (c) average flight time. (d) UAV navigation trajectory.

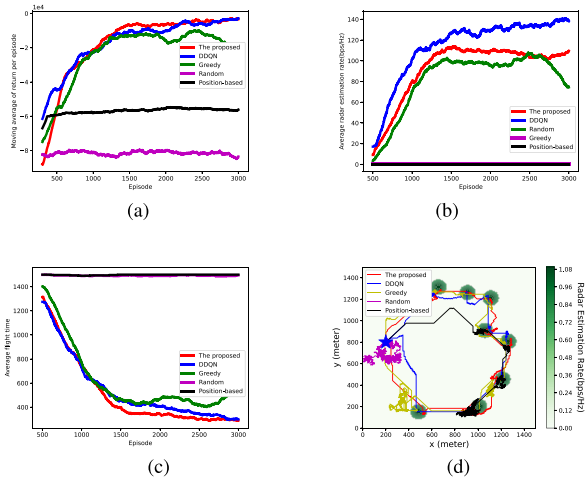


Fig. 11. Performance comparison versus different algorithms under $C_{th} = 10$ bps/Hz. (a) Moving average returns. (b) average radar estimation rate. (c) average flight time. (d) UAV navigation trajectory.

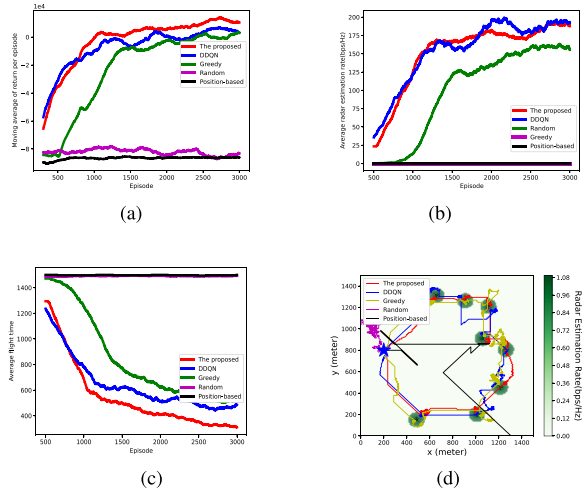


Fig. 12. Performance comparison versus different algorithms under $C_{th} = 20$ bps/Hz. (a) Moving average returns. (b) average radar estimation rate. (c) average flight time. (d) UAV navigation trajectory.

the direct flight method approaches the sensing targets based on the received target signal power strength and returns to the initial location $\mathbf{q}_I$ based on the distances between the UAV and $\mathbf{q}_I$. Fig. 9, which is based on Fig. 6, includes the average radar estimation rate of the direct flight method for easy comparison. As shown in Fig. 9, the average radar estimation rate of the proposed algorithm is consistently higher than that of the direct flight method.

The performance comparison of the proposed algorithm with RL methods is presented as follows. Firstly, we analyze the convergence of the average moving returns of different algorithms under different values of C_{th}. From Figs. 10(a), 11(a), and 12(a), it can be observed that the average moving returns of the proposed algorithm and DDQN algorithms achieve stable convergence. Moreover, the convergence speed and convergence values of the proposed algorithm are consistently better than those of the DDQN algorithm. On the other hand, the Greedy algorithm shows an unstable convergence effect as the value of C_{th} increases. This is due to the fact that the Greedy algorithm only considers the currently optimal trajectory, which may not be globally optimal when there is insufficient knowledge of the environment. Furthermore, it can be observed that the Random algorithm does not show any convergence trend, and the convergence of the Position-based algorithm is highly unstable. In fact, the average moving return of the Position-based algorithm are even lower than that of the Random algorithm when $C_{th} = 20$ bps/Hz. This poor convergence performance of the Position-based algorithm is attributed to the fact that the UAV receives different rewards during different sensing target phases even at the same position, resulting in ineffective UAV navigation when using position alone as a state input. It is worth noting that the proposed algorithm consistently outperforms the other four algorithms in terms of convergence performance under different values of C_{th}. This indicates that the proposed algorithm exhibits stable and efficient convergence performance across different requirements of C_{th}.

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The optimization goal of this paper is to minimize the UAV flight time and maximize the system radar estimation rate, while ensuring that the UAV trajectory satisfies the constraints (29b)–(29g). Subsequently, we analyze the flight time and the system radar estimation rate of different algorithms when the trajectory constraints are satisfied. From Figs. 10(d), 11(d), and 12(d), it can be observed that both the proposed algorithm and the DDQN algorithm can successfully navigate the UAV trajectory while satisfying the trajectory constraints after training. However, the effect of the Greedy algorithm is highly unstable in achieving UAV navigation. Moreover, from Figs. 10(c), 11(c), and 12(c), it can be seen that the average flight time of the proposed algorithm is lower than that of both the DDQN algorithm and the Greedy algorithm under different values of C_{th}. According to Figs. 10(b), 11(b), and 12(b), the proposed algorithm achieves more stable convergence values of radar estimation rate compared to the DDQN and Greedy algorithms. It can also be visually observed from Figs. 10(d), 11(d), and 12(d) that the Random and Position-based algorithms are unable to implement UAV navigation while satisfying the trajectory constraints. Therefore, the average radar estimation rate and average flight time of Random and Position-based algorithms are consistently N_{max} and 0 bps/Hz, respectively. Accordingly, we can conclude that the proposed algorithm exhibits more efficient and stable performance compared to the other benchmark methods.

VI. Conclusion

In this article, we investigate a UAV-assisted IoT system with ISAC capability, where the UAV trajectory is designed to optimize the weighted sum of UAV flight time and the system total radar estimation rate, while ensuring sensing all IoT nodes. We propose an advanced multi-step learning Dueling DDQN algorithm to solve the optimization problem. By showing the performance of the proposed algorithm under various requirements for radar estimation rate, we demonstrate that the proposed algorithm exhibits stable and efficient convergence performance, consistently achieving the goal of maximizing the system radar estimation rate while minimizing UAV flight time. Furthermore, we substantiate the algorithm's superiority through a comparative analysis with benchmark methods, including the direct flight method, DDQN algorithm, Position-based algorithm, Greedy algorithm, and random algorithm.

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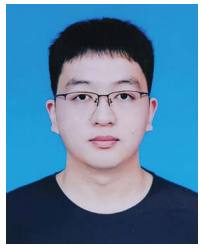
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